

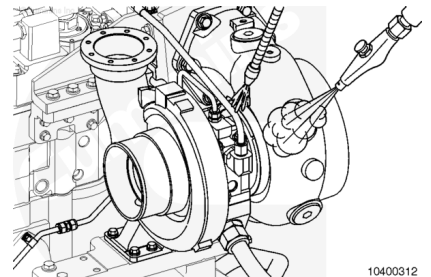
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

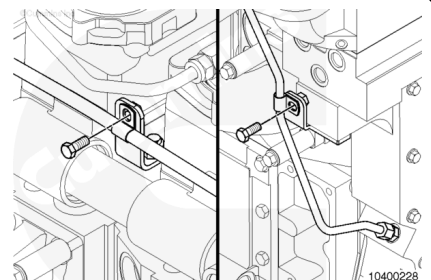
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

- Clean the area around the turbocharger.



Remove

Remove the capscrews securing the turbocharger oil supply tube clips to the braces.

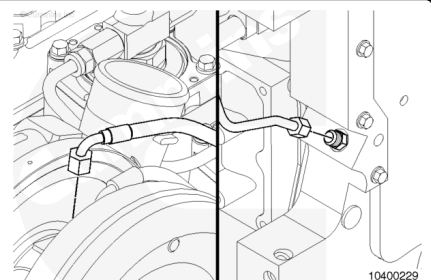


⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



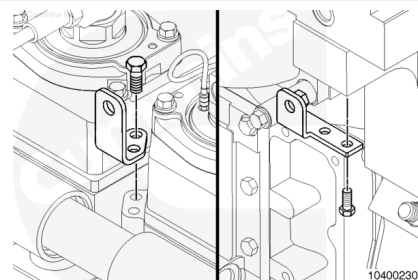
Disconnect the turbocharger oil supply hose line from the turbocharger.

Remove the male union from the turbocharger and discard the o-rings.

Disconnect the turbocharger oil supply tube from the engine.

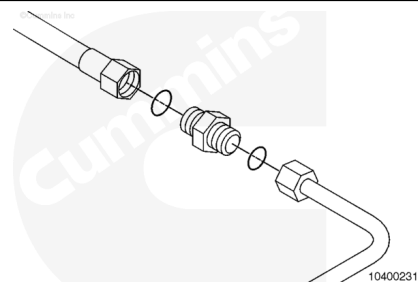
Remove the male union and discard the o-rings.

Remove the capscrews securing the turbocharger oil supply line braces to the engine.



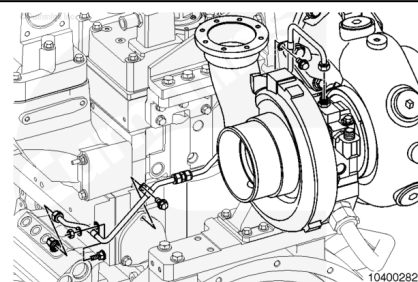
Disconnect the turbocharger oil supply hose from the turbocharger oil supply tube at the male union.

Discard the o-rings from the male union.



Marine Applications

Remove the p-clips and turbocharger oil supply line.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the

possibility of personal injury.

⚠ WARNING ⚠

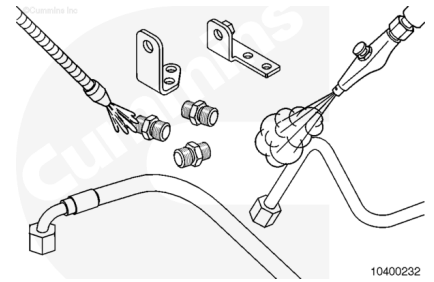
Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

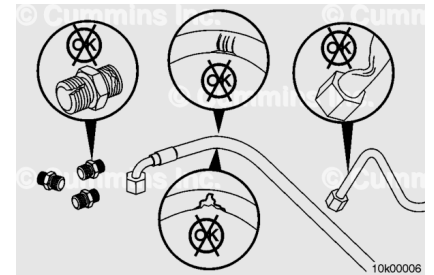
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean all components with solvent.

Dry with compressed air.



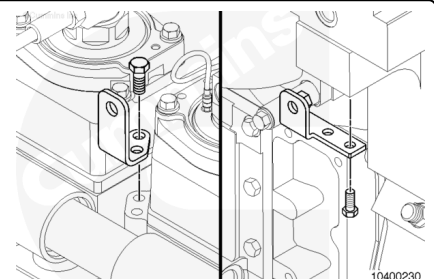
- Inspect the turbocharger oil supply hose for damage, kinks, fretting, or restrictions.
- Inspect the turbocharger oil supply tube for damage, kinks, or restrictions. Replace the tube, if damaged.
- Inspect and replace any of the clips, if damaged or worn.
- Inspect the male unions for damage, including stripped threads. Replace male unions, if damaged .
- Inspect the oil supply line braces for cracks, other damage, and corrosion. Replace any braces that are damaged.



Install

Install the turbocharger oil supply line braces with capscrews.

Torque Value: 40 n•m [30 ft-lb]

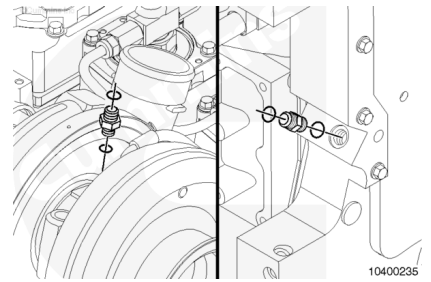


Install new o-rings on the male union and install on the turbocharger.

Torque Value: 25 n•m [221 in-lb]

Install new o-rings on the male union and install on the engine.

Torque Value: 25 n•m [221 in-lb]



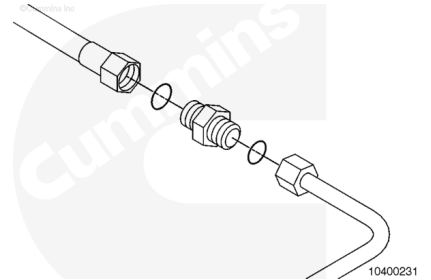
Install a new o-ring on the male union and attach the turbocharger oil supply hose.

Torque Value: 25 n•m [221 in-lb]

If more clearance is needed for the turbocharger oil supply line, there is a 1 inch extended male union available for the turbocharger oil supply line.

Install a new o-ring on male union and attach the turbocharger oil supply tube.

Torque Value: 25 n•m [221 in-lb]



Install the turbocharger oil supply line to the turbocharger male union.

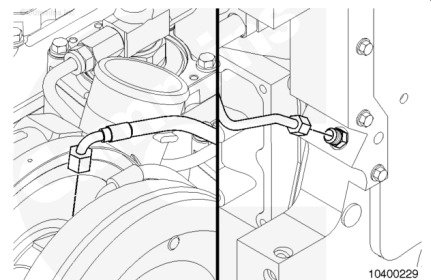
Torque Value: 25 n•m [221 in-lb]

Make sure that the turbocharger oil supply line is **not** in contact with the coolant return hose. If they are in contact and more clearance is needed, there is a 1 inch extended male union available for the turbocharger oil supply line.

Install the turbocharger oil supply line to the engine male union.

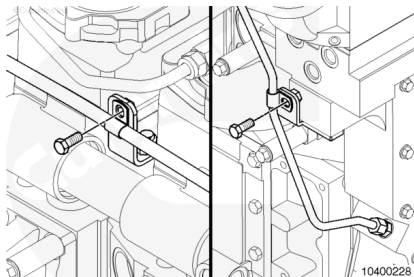
Torque Value: 25 n•m [221 in-lb]

Use the following procedures for proper speed sensor wire routing if the 1 inch extended male union is upfitted on an engine with a turbocharger speed sensor. Use the following procedure in the QSK19 (CM850 Modular Common Rail System) Electronic Control System Troubleshooting and Repair Manual, Bulletin 4021493. Refer to Procedure 019-390 in Section 19 (</qs3/pubsys2/xml/en/procedures/13/13-019-390.html>)



Attach the turbocharger oil supply tube with P-clips and capscrews.

Torque Value: 40 n•m [30 ft-lb]

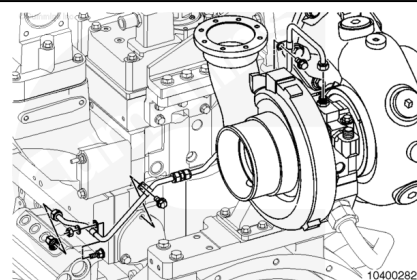


Marine Applications

Install the turbocharger oil supply tube. These line connections are designed to be tightened until bottomed out.

Install the P-clips.

Torque Value: 40 n•m [30 ft-lb]



Finishing Steps

Operate the engine and check for leaks.

